MS^{2}ANet: A Multiscale Spatial-Spectrum Agent Attention Network for Crop Hyperspectral Image Unmixing

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***Abstract*—Hyperspectral image (HSI) analysis often faces the challenge of pixel mixing, primarily due to limitations in spectrometer resolution, particularly when dealing with the complex and dynamic surface features of natural environments. As societal demands increase, the complexity of surface information has also risen, especially in agricultural scenarios, where high-density crop growth and the mixing of spectral features at boundaries make HSI unmixing more challenging. However, existing methods have struggled to effectively capture spatial information, especially the boundary information between crops and the background. To address these challenges, this article introduces a novel crop unmixing algorithm that leverages multiscale and spatial–spectral attention mechanisms. The algorithm employs a multiscale dilated convolution module that aggregates local spatial features and contextual information at different scales, providing multiscale spatial cues for subsequent attention modules and improving the characterization of mixed pixels near crop boundaries. In addition, by integrating a spatial-spectrum agent attention mechanism (S^{2}AM), the model dynamically adjusts and strengthens both global spatial and spectral information, significantly improving its focus on critical regions. The inclusion of this agent mechanism further reduces computational complexity, enhancing the model's efficiency in complex crop scenario data. Moreover, the triplet multilayer perceptron (MLP) module enables MS^{2}ANet to effectively encode and integrate features across height, width, and spatial dimensions. Extensive experiments demonstrate the strong performance of MS^{2}Anet on two crop hyperspectral unmixing datasets, as well as on the publicly available Samson and Apex datasets, validating the robustness and general applicability of the method.**

***Index Terms*—Crop hyperspectral image (HSI) unmixing, multiscale dilated convolution, spatial-spectrum agent attention, transformer.**

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I. Introduction

HYPERSPECTRAL remote sensing technology, with its high spectral resolution and integrated spectral imaging capabilities, has been widely applied in fields such as precision agriculture, environmental monitoring, and mineral exploration. In recent years, hyperspectral remote sensing imagery has gradually become an essential data source for quantitative remote sensing research in crop nutrition diagnosis, disease early warning, and phenotyping [1]. Hyperspectral unmixing, with its ability to separate endmember spectra and abundance information from mixed pixels, has been extensively applied in the precise extraction of crop planting areas, among other areas. However, traditional image segmentation methods often struggle to accurately distinguish boundaries between crops or between crop and noncrop regions, leading to significant errors in the extraction results. With the continuous advancement of computer and remote sensing technologies, hyperspectral unmixing techniques have made crop area extraction more accurate. However, due to the complexity of terrestrial surfaces and the spatial resolution limitations of hyperspectral imagers, each pixel in hyperspectral imagery often contains multiple spectral features, creating a phenomenon known as "mixed pixels," which poses significant challenges for remote sensing image interpretation.

Therefore, hyperspectral unmixing for crops, in order to obtain spectral reflectance information of materials, is a critical

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prerequisite for applying hyperspectral unmixing in the precise extraction of crop planting areas and the fine monitoring of soil quality in complex land-cover scenarios. Hyperspectral unmixing can be regarded as a special case of blind source separation in machine learning, where the spectral signals of pixels in hyperspectral imagery are decomposed into a series of material-specific spectral signals and their corresponding abundance coefficients under spectral mixing conditions [2].

Currently, the linear spectral mixing model serves as the foundational method for hyperspectral unmixing, widely applied due to its simplicity and efficiency. However, it still faces limitations in handling spectral nonlinearity, noise interference, and spectral variability, which affect the accuracy and robustness of unmixing results. Meanwhile, although nonlinear unmixing models show considerable potential in handling complex mixing relationships, their high model complexity, poor physical interpretability, and heavy reliance on data limit their widespread application. Future research may focus on combining deep learning with traditional linear unmixing models to complement the strengths of both approaches. In the past two decades, a rich family of traditional hyperspectral unmixing methods has been developed before the advent of deep learning. Classical geometrical approaches, such as N-FINDR, vertex component analysis (VCA), and minimum-volume-based algorithms, estimate endmembers by exploiting the convex geometry of the data cloud [3], [4]. Statistical and sparse-regression-based methods further model endmember distributions or impose sparsity priors on abundances [2], [5]. In addition, fully constrained least squares (FCLSs) and nonnegative matrix factorization (NMF)-based formulations explicitly enforce the abundance nonnegativity and sum-to-one constraints, and have been widely adopted as baselines in practical applications [6]. These traditional methods provide physically interpretable, computationally efficient baselines, and many recent deep learning approaches are designed to inherit or extend their modeling assumptions.

From the perspective of deep learning, existing linear unsupervised hyperspectral unmixing methods can be broadly grouped into three categories.

1) Autoencoder (AE)-based methods [7], [8], [9], [10], [11], including denoising AEs, sparse and variational AEs (VAEs), learn low-dimensional abundance codes while reconstructing endmember spectra.

2) Dual-driven data-model unmixing methods [12], [13], [14], such as SNMF-Net, ADMM-based unmixing networks, and EGU-Net, unfold classical optimization or physical linear mixing models (LMMs) into interpretable deep architectures.

3) Hybrid methods that combine traditional models with deep networks [15], [16], [17], e.g., sparsity-constrained deep NMF (DNMF) and AE-like NMF, which enhance the flexibility of NMF while preserving physical interpretability.

Beyond AE-like architectures, recent works also explore convolutional, generative, and transformer-based frameworks. Deep spectral convolutional network (DSCN++) and 3-D CNNs explicitly model local spatial–spectral structures [18], [19]; deep generative models capture endmember variability and uncertainty [20], [21]; and transformer-based unmixing networks leverage self-attention to learn long-range spectral–spatial dependencies [22], [23]. These representative deep learning paradigms provide diverse yet complementary baselines, some of which are adopted as strong competitors in our experiments.

In the subsequent paragraphs, we review and summarize these three categories of existing deep learning-based linear unsupervised unmixing methods. AE-based hyperspectral unmixing methods generate low-dimensional representations through the encoder to obtain abundance information while utilizing the weights between nodes in the decoder to extract endmember spectra. These methods are based on unsupervised learning and effectively separate endmembers and abundances from hyperspectral data, becoming an important branch of unsupervised unmixing methods. Several related studies are reviewed as follows. Guo et al. [24] first proposed a hyperspectral unmixing method based on AE cascades, introducing AEs into hyperspectral unmixing tasks. This method consists of two parts: an edge denoising AE and a nonnegative sparse AE stack. The denoising AE removes noise from hyperspectral data, and the nonnegative sparse AE performs blind signal separation to extract endmember information and abundances. While the method performs well in noise reduction, it does not fully account for spectral variability and complex noise influences, leading to degraded unmixing performance under conditions with spectral variability or high noise. Palsson et al. [19] proposed a hyperspectral unmixing method based on a neural network AE, using custom layers and weight constraints to achieve nonnegativity constraints (ANCs) and abundance sum-to-one constraints (ASCs). Meanwhile, spectral angle distance (SAD) is adopted as the objective function to enhance the model's adaptability to spectral variability. Qu and Qi [25] designed an unsupervised sparse decoupled denoising AE (uDAS), overcoming the limitations of weight sharing in traditional AEs through decoupled design. This method introduces norm constraints in the encoder to reduce the impact of redundant endmembers while integrating denoising capabilities into network optimization to avoid additional reconstruction errors. This design demonstrates excellent endmember extraction and abundance estimation performance in high-noise data. Su et al. [26] proposed a hyperspectral unmixing method based on stacked nonnegative sparse AEs, which involves two stages: the first stage uses nonnegative sparse AEs to detect and correct outliers in hyperspectral data, and the second stage uses a final AE to extract endmember spectra and abundances. This method significantly improves unmixing robustness by introducing sparsity and nonnegativity constraints, especially in scenarios with outliers and low signal-to-noise ratios. Su et al. [27] further explored a hyperspectral unmixing method based on a deep AE network (DAEN). This method uses a stacked AE to initialize the unmixing network, generates initial unmixing parameters through spectral feature learning, and then utilizes VAEs for blind signal separation to extract endmembers and abundances. This method shows high accuracy and adaptability in handling complex spectral scenarios.

Furthermore, some studies have specifically addressed spectral variability and achieved good results. For instance, Ozkan and Akar [18] proposed an improved DSCN++, which

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incorporates trainable uncertainty terms and generative adversarial network (GAN) optimization strategies to capture endmember uncertainty and spectral variability, significantly mitigating the impact of spectral variability on unmixing performance. In the same year, Palsson et al. [19] proposed an unmixing method based on an improved 3-D convolutional neural network (3D-CNN), which combines principal component analysis (PCA) for dimensionality reduction and 3-D convolutional feature extraction to reduce the interference of anomalous endmembers and noise in unmixing results. Borsoi et al. [20] designed a deep generative endmember modeling method, using a VAE to construct a low-dimensional generative model for endmember spectra while embedding physical mixing models to capture spectral variability, thus overcoming the lack of modeling flexibility in traditional models. Dou et al. [28] proposed an AE method based on orthogonal sparse priors, combining orthogonal sparse priors, super-Laplacian loss functions, and data-driven outlier detection mechanisms to effectively mitigate issues related to outliers, complex noise, and the inapplicability of traditional Gaussian error models. Shi et al. [21] proposed a hyperspectral unmixing method based on probabilistic generative models (PGMSU), modeling endmember variability with deep generative models and using variational inference (VI) to estimate endmember distributions and abundances. This method overcomes the limitations of traditional LMMs in handling complex endmember variability.

The dual-driven data-model unmixing method is a hyperspectral unmixing technique that combines both data-driven and model-driven strategies. Compared to purely AE-based unmixing methods, these methods not only rely on deep neural networks to learn prior knowledge from data but also incorporate physical model constraints and inversion algorithms, combining physical interpretability with powerful learning capabilities. In this framework, the data-driven part extracts features and latent patterns from hyperspectral imagery through deep learning techniques, while the model-driven part provides theoretical guidance for the unmixing process by embedding constraints and inversion algorithms based on physical mixing models (e.g., LMMs). This combination overcomes the limitations of single-driven approaches and demonstrates better performance in scenarios with significant noise interference, spectral variability, or limited training data. Qian et al. [29] proposed a model-heuristic neural network method that uses a model-heuristic module as a feedforward process, iterating based on iterative shrinkage thresholding algorithms (ISTAs) and LMMs to expand the iterative process into a hierarchical structure. This approach addresses the lack of physical interpretability and the need for large amounts of training data in traditional neural networks for hyperspectral unmixing tasks. Zhou and Rodrigues [13] proposed a hyperspectral unmixing network method based on the alternating direction method of multipliers (ADMMs), which unfolds the ADMM solver's iterative process into a hierarchical structure within a neural network. This method combines sparse-driven hyperspectral linear unmixing models to construct the neural network, realizing a data- and model-driven joint unmixing process. This method effectively combines the advantages of model-driven and data-driven approaches, improving unmixing accuracy, especially in scenarios with limited training data, achieving rapid convergence, and providing good unmixing performance.

Unlike the aforementioned two methods, the traditional-deep combined unmixing approach is based on traditional mathematical models, defining the basic framework for the unmixing problem and optimizing the unmixing process with deep learning models. The goal is to use deep learning's powerful nonlinear feature extraction ability to compensate for the limitations of traditional methods in handling complex real-world data while maintaining the physical interpretability and theoretical constraints of traditional methods. Feng et al. [30] proposed a sparse constraint method combining DNMF with total variation (TV), integrating traditional NMF with a deep learning framework. By introducing deep learning ideas, the traditional NMF algorithm is extended into a DNMF model. The model extracts richer spectral information during the pretraining phase of hierarchical features and adjusts all factors during fine-tuning to reduce total reconstruction error. Sparse constraints are added during optimization, and TV regularization is applied to maintain spatial smoothness in endmember abundances. This method effectively overcomes the flexibility limitations of traditional NMF methods when handling complex data while compensating for the shortcomings of deep learning methods in dealing with spectral data sparsity. However, due to the weak robustness of traditional NMF methods in noisy environments and their insufficient utilization of spatial context information, Feng et al. [15] later proposed a class AE NMF (CANMF) method based on mutual entropy. This method enhances model robustness to noise and effectively handles hyperspectral data in high-noise environments, further optimizing the spatial distribution of endmember abundances and maintaining smoothness in local areas. In the same year, Rasti et al. [31] proposed the minimum simplex convolutional network (MiSiCNet), which further combines geometric and spatial information to achieve efficient hyperspectral unmixing. Using a deep encoder–decoder network structure, they incorporated geometric information into endmember and abundance estimation, designing a geometric penalty function. This method effectively solves the shortcomings of traditional LMMs in capturing geometric features and handling nonpure pixel scenarios.

Currently, researchers worldwide have conducted extensive studies on hyperspectral unmixing for common crops and have proposed a variety of effective methods. For example, Adams from the University of Colorado first applied hyperspectral unmixing technology to soil composition analysis in 1993 [32]. The study demonstrated that hyperspectral unmixing could effectively separate different components in soil spectra, enabling accurate estimation of soil organic carbon content. Yao et al. [33] from the Remote Sensing and GIS Institute at Peking University summarized the applications of hyperspectral unmixing technology in agriculture, including studies on crop leaf spectral characteristics and the inversion and extraction of crop ecological and physical parameters. Chi [34] from the University of California, focusing on

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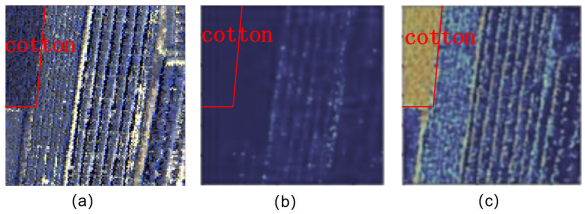


Fig. 1. (a) Cotton region in the HSI. (b) Results of the traditional crop unmixing algorithm. (c) Actual cotton abundance map.

agricultural fields, explored the use of spectral features for quantitatively evaluating crop residues. Paura and Marcinkevičius [35] from Vilnius University in Lithuania created a new hyperspectral blueberry crop unmixing dataset and achieved good unmixing results using an improved U-Net model on this dataset. Pan et al. [36] proposed a method for estimating green algae area based on spectral unmixing. By using endmember extraction algorithms, they automatically identified the spectral features of green algae in images and combined the original hyperspectral data with an FCLSs method to compute the abundance map of green algae and estimate its coverage area. This method effectively addressed the issue of inaccurate area estimation due to the resolution limitations of hyperspectral images (HSIs). Shilong et al. [37] proposed a hyperspectral unmixing method for rice fields that combines UAV high-resolution images with hyperspectral remote sensing imagery. They used spatial registration technology to align high-resolution digital orthophotos with UAV HSIs and then performed object classification using an SVM classifier. After matching the classification results with hyperspectral pixels, they computed the endmember abundances for each pixel. This method overcame the limitations of single data sources, enhanced the richness of the data, and improved the accuracy of endmember abundance calculations for rice field pixels. Hu et al. [38] addressed issues of mixed pixels and endmember spectral variability in wheat planting area inversion by proposing a self-supervised learning-based spectral unmixing (SLSU) algorithm. This method uses VAEs to model endmember spectral variability and generate endmember libraries in an unsupervised manner, combines alternating least squares and FCLSs for endmember abundance estimation, and uses probabilistic relaxation labeling for spatial neighborhood correction.

Existing hyperspectral unmixing methods for crops mainly rely on traditional feature extraction algorithms. Although these algorithms perform reliably when processing general image data, they often fail to fully exploit the rich spectral information inherent in hyperspectral imagery when applied to hyperspectral data. More importantly, these traditional methods significantly lack the ability to utilize spatial information, preventing them from effectively exploiting the spatial continuity and correlation within HSIs. This inadequacy in the combined application of spectral and spatial information typically leads to inaccurate unmixing results and reduced model generalization.

In crop hyperspectral unmixing, a large portion of mixed pixels is located near field boundaries or in regions where crops, soil, and background objects coexist. Existing deep unmixing networks typically incorporate spatial information via standard convolutions or global self-attention, but they either under-exploit multiscale boundary cues or suffer from prohibitive computational costs at high spatial resolutions. In addition, most MLP-based designs focus on spectral or channel-wise mixing and rarely encode height, width, and channel dimensions jointly. These limitations motivate us to develop an architecture that can aggregate multiscale local context around crop boundaries, efficiently model nonlocal spatial relationships and global spectral correlations via agent attention, and enhance pixel-level detail representation through a triplet MLP operating along height, width, and channel dimensions. For instance, in Fig. 1, the traditional unmixing method fails to clearly depict the abundance of the cotton-growing area and does not distinguish boundaries between different crop planting regions. To address the issue where existing hyperspectral unmixing methods for crops do not effectively capture spatial information, especially the key boundary features, this article proposes a novel crop unmixing algorithm based on a multiscale and spatial–spectral attention mechanism. The algorithm first employs a multiscale dilated convolution module to aggregate local spatial features and contextual information at different scales, providing multiscale spatial cues for subsequent modules. In addition, by incorporating a spatial-spectrum agent attention mechanism (S^2AM), the algorithm dynamically weights and enhances both global spatial and spectral information, so that these multiscale cues are jointly exploited to strengthen the model's focus on mixed pixels and regions near crop boundaries. The introduction of the agent mechanism further reduces the computational complexity of the model, improving the applicability and computational efficiency of the unmixing algorithm in complex crop scenario data.

The key contributions of the proposed method are outlined as follows.

1) To enhance the unmixing capability for complex land-cover scenes, a novel unmixing method that integrates multiscale expansion convolution with spatial-spectrum agent attention is proposed. Initially, a multiscale dilated convolution module is designed to aggregate spatial information at multiple receptive-field sizes. By jointly modeling fine-scale details and broader contextual cues, this module provides richer spatial features for the subsequent spatial-spectrum agent attention module, which improves the discrimination of mixed pixels near crop boundaries.
2) To efficiently capture global spatial characteristics and spectral information, we introduce a novel spatial-spectrum attention module. This module not only captures nonlocal context feature dependencies but also identifies global spectral correlations, thereby enhancing the model's capability to focus on global regions. In addition, the incorporation of an agent mechanism significantly reduces computational complexity, improving both the applicability and computational efficiency of the hybrid algorithm for complex crop scene data.
3) To more effectively capture hyperspectral pixel-level details and aggregate multidimensional encoded

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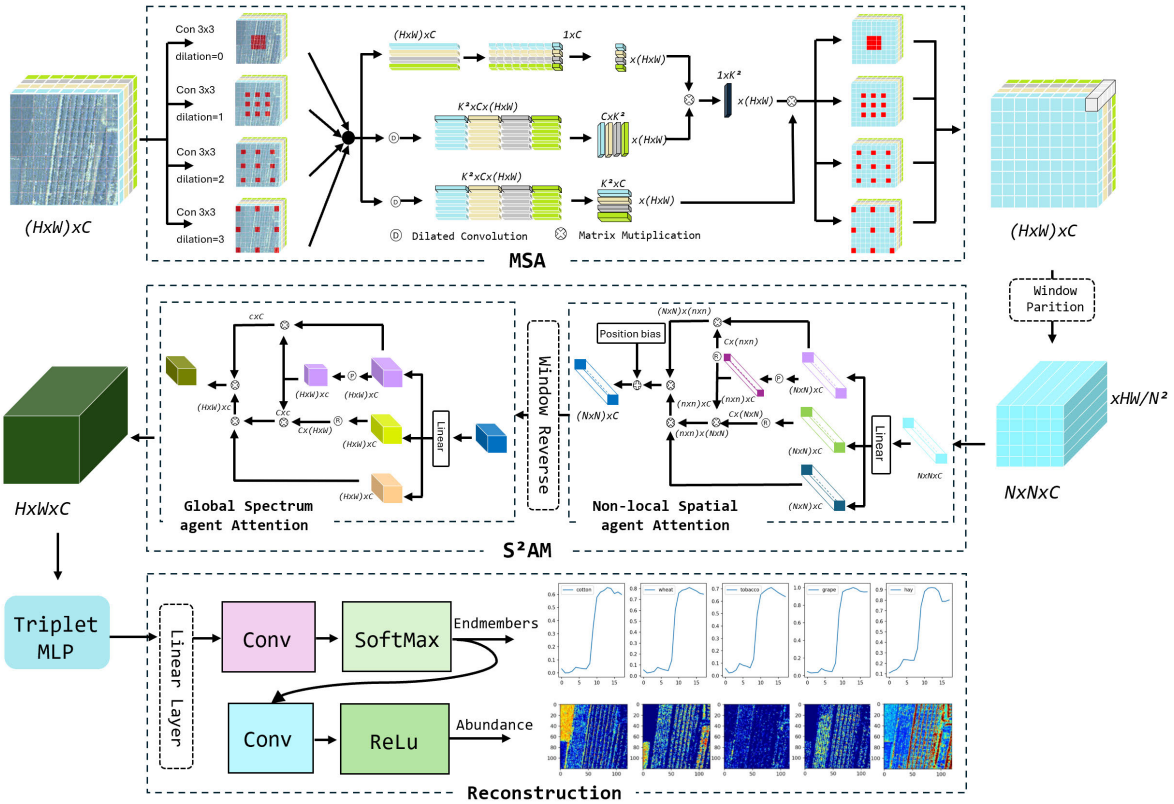


Fig. 2. This article presents the MS²ANet architecture, which is specifically designed for the separation of HSI in agricultural applications. The core structure of MS²ANet comprises two primary components: an encoder and a decoder. The encoder module incorporates an MSA, an S²AM, and a triplet-based MLP model.

features, we propose a novel triadic multilayer perceptron (MLP) module. This module not only encodes feature representations in the height and width dimensions but also captures diverse spatial information across multiple spatial dimensions. By integrating these multidimensional features, the model's capability to identify target substances is significantly enhanced, thereby improving the accuracy of spectral unmixing.

The structure of this article is organized as follows. Section II discusses the proposed algorithm for HSI unmixing. Section III applies the methodology to four real datasets; two of them are specially made for crop hyperspectral unmixing datasets, and the experimental results are fully analyzed and discussed. Finally, comprehensive conclusions are drawn in Section IV.

II. Proposed Method

The linear spectral mixing model, known for its structural simplicity and broad applicability, is extensively utilized in unmixing processes. In this model, the formula representing the spectral reflectance observed by a sensor, resulting from the interaction of incident light with a specific surface material, is given by

$$Y = \sum_{j=1}^{P} e_j a_j + N \tag{1}$$

$$a_j \geq 0, \quad \sum_{j=1}^{P} a_j = 1 \tag{2}$$

where the matrix Y represents the HSI, P represents the number of endmembers present in the HSI, e denotes the endmembers, a represents the abundances, and N accounts for additional noise present in Y. The abundance matrix adheres to two physical constraints: the ANC and the ASC.

Fig. 2 illustrates the unmixing model based on the multiscale S²AM, which consists of two main modules: encoding and decoding. The spectral unmixing module includes a multiscale dilated convolution module, a spatial-spectrum agent attention module, and a triaxial MLP module, while the decoding module comprises a fully connected layer and two convolutional layers. The diagram shows that the input HSI is processed through the multiscale dilated convolution module to capture local spatial features from the surrounding area. By aggregating features from different spatial resolutions of the land-cover types, this module provides multiscale spatial cues that are further exploited by the spatial-spectrum agent attention module to identify and differentiate materials in mixed pixels at finer scales. The spatial-spectrum agent attention module not only computes correlations along the spatial dimension and captures long-range dependencies between different locations in the sequence but also calculates the attention weights of spectral vectors to capture correlations across spectral sequences [39]. Following this, the triaxial

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MLP module encodes the feature representations across the height, width, and spatial dimensions, thereby enhancing the model's ability to recognize target materials. The output of the fully connected layer is reshaped to match the size of the abundance map, while the convolutional layers are used to reduce noise. In addition, the SoftMax activation function is employed to enforce ASC and ANC, resulting in the final abundance map [40]. Finally, the decoder utilizes a single convolutional layer with endmember weighting to increase the number of channels to match the number of bands in the HSI. In the following sections, Section II-A describes the multiscale dilated convolution module, Section II-B outlines the spatial-spectrum agent attention module, Section II-C discusses the triaxial MLP module, and Section II-D details the decoding module.

A. Multiscale Dilated Convolution Module for Enhanced Spectral Analysis in Deep Learning

Multiscale approaches in deep learning have gained widespread application in various fields, such as computer vision and remote sensing image analysis, due to their ability to capture features at different resolution levels [22], [41], [42], [43]. In the context of complex crop HSIs, the spatial features of different land-cover types may overlap or mix at the boundary regions, causing a single pixel to contain multiple crop features. In relatively small areas, crop boundary characteristics are difficult to distinguish. By aggregating spatial feature information across different scales, the module can identify and capture small changes and differences, which are then utilized by the spatial-spectrum agent attention module to better handle mixed pixels near crop edges and improve the accuracy of crop unmixing.

The multiscale dilated convolution module proposed in this article first passes the input image through four expansion convolution modules in parallel. In practice, we adopt dilation rates 0, 1, 2, 3 for the four parallel branches. As shown in Section III-E, this configuration strikes a good balance between preserving fine local structures and capturing broader contextual information, and yields the best overall unmixing accuracy among the tested combinations. The module operates by setting query ($\boldsymbol{Q_0}$), key ($\boldsymbol{K_0}$), and value ($\boldsymbol{V_0}$) as the three inputs for calculating attention weights. Specifically, Q_0, representing the features at the current location, is used to compute the relevance with features at other locations. $\boldsymbol{K_0}$ denotes the feature vectors from other locations, which are used to calculate their relevance with the query. $\boldsymbol{V_0}$ represents the feature vectors at other locations and provides weighted contextual information to the query based on relevance. Unlike the transformer model [44], both $\boldsymbol{K_0}$ and $\boldsymbol{V_0}$ in the dilated convolution attention module are derived through dilated convolutions, capturing semantic information within the expanded receptive field. The parallel arrangement of dilated convolution attention modules allows the model to capture features of land cover at different resolutions and spectral heterogeneity across spatial scales. By acquiring a broader context, the model's performance is significantly enhanced. The operational flow of this module is detailed in Fig. 3, and the algorithmic steps involved in the dilated convolution attention mechanism are as follows.

1) Assuming the input tensor has a shape of $X_0 \in R^{B\times H\times W\times C}$, perform transformation operations on $\boldsymbol{Q_0}$, and execute dilated convolution and transformation operations on $\boldsymbol{K_0}$ and $\boldsymbol{V_0}$, resulting in $\boldsymbol{Q_0} \in R^{B\times C\times L_q}$, $\boldsymbol{K_0} \in R^{B\times C\times K^2\times L_k}$, and $\boldsymbol{V_0} \in R^{B\times K^2\times C\times L_v}$

$$\boldsymbol{Q_0} = \text{query_transform}\,(X_0) \in R^{B\times C\times L_q} \tag{3}$$

$$\boldsymbol{K_0} = \text{key_transform}\,(\text{dilated_conv}\,(X_0)) \in R^{B\times C\times K^2\times L_k} \tag{4}$$

$$\boldsymbol{V_0} = \text{value_transform}\,(\text{dilated_conv}\,(X_0)) \in R^{B\times K^2\times C\times L_v}. \tag{5}$$

   In this context, L_q, L_k, and L_v represent the sequence lengths of $\boldsymbol{Q_0}$, $\boldsymbol{K_0}$, and $\boldsymbol{V_0}$, respectively, with $L_q = L_k = L_v = H \times W$. In addition, the term K^2 refers to the expansion by a factor $K \times K$ of the input tensor X after dilation padding, using a dilation convolution kernel of size K^2.

2) Then, attention weights $D \in R^{B\times 1\times K^2\times H\times W}$ are computed through matrix multiplication and softmax operations between $\boldsymbol{Q_0}$ and $\boldsymbol{K_0}$, reflecting the similarity between $\boldsymbol{Q_0}$ and $\boldsymbol{K_0}$

$$D = \text{softmax}\,(\boldsymbol{Q_0} \otimes \boldsymbol{K_0}) \in R^{B\times 1\times K^2\times H\times W} \tag{6}$$

   where $\otimes$ denotes matrix multiplication.

3) Finally, the output tensor is obtained by multiplying the attention weights D with the matrix $\boldsymbol{V_0}$ and adding the result to the input tensor to produce the final attention output Attention $\in R^{B\times H\times W\times C}$

$$\text{Attention} = (\text{transform}\,(D \otimes \boldsymbol{V_0}) \oplus X_0) \in \mathbf{R}^{B\times H\times W\times C} \tag{7}$$

   where $\otimes$ denotes matrix multiplication, $\oplus$ represents matrix addition, and the transform function refers to the transformation operation on matrices.

B. Spatial-Spectrum Agent Attention Module

To address the inherent characteristics of hyperspectral imagery, namely, strong nonlocal spatial correlations among pixels of the same land-cover type and high spectral redundancy across hundreds of continuous narrow bands, this article proposes a customized spatial-spectrum agent attention module. Different from the generic self-attention mechanism commonly used in transformers, this module is explicitly designed to leverage the structural properties of HSI while alleviating its computational bottlenecks. The module consists of two complementary submodules: nonlocal spatial agent attention (NSA) and global spectral agent attention (GSA). Both introduce compact agent vectors as information hubs to avoid the prohibitive computational cost of dense pixel-to-pixel and band-to-band similarity calculations required by traditional self-attention.

In the spatial domain, since the same-type crops in agricultural scenes are often scattered in disjoint parcels with strong nonlocal spatial dependencies, traditional self-attention

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is impractical due to excessive computational cost. The NSA aggregates global spatial correlations via compact agent vectors and feeds them back to local windows to capture global context efficiently; paired with a multiscale dilated convolution submodule that enhances local feature perception, it enables the model to both identify crop boundaries accurately and model cross-parcel spatial associations fully. In the spectral domain, targeting severe multicollinearity and redundancy in HSI bands, the GSA projects high-dimensional spectral sequences into a low-dimensional agent space via spectral agent vectors. As an information bottleneck, these agents distill redundant bands into a small set of core independent spectral signatures before attention map reconstruction, effectively capturing long-range spectral dependencies. The execution process of the spatial-spectrum agent module is as follows.

1) Assuming the input $X_1 \in R^{B\times H\times W\times C}$, the input tensor is divided into HW/N^2 sequence blocks of size $N \times N \times C$, according to the spatial dimension. These sequence blocks are then treated as the inputs for $\boldsymbol{Q_1}$, $\boldsymbol{K_1}$, and $\boldsymbol{V_1}$ in the attention mechanism. Importantly, the interpretations of $\boldsymbol{Q_1}$, $\boldsymbol{K_1}$, and $\boldsymbol{V_1}$ align with their roles in the multiscale dilated convolution module, despite variations in the computation process and dimensions of the resulting tensors. The specific computation process of the NSA module is detailed in (8)–(15), and the calculation process of the NSA module is described in (16)–(23)

$$\boldsymbol{Q_1} = \text{query_parition}\,(X_1) \in R^{N\times N\times C\times L_Q} \tag{8}$$
$$\boldsymbol{K_1} = \text{key_parition}\,(X_1) \in R^{N\times N\times C\times L_K} \tag{9}$$
$$\boldsymbol{V_1} = \text{value_parition}\,(X_1) \in R^{N\times N\times C\times L_V} \tag{10}$$

where $L_Q = L_K = L_V = HW/N^2$, representing the number of sequence blocks.

2) Differing from traditional transformers, the method avoids direct computation of pairwise similarities between $\boldsymbol{Q_1}$ and $\boldsymbol{K_1}$, opting instead to use a small set of agent vectors $\boldsymbol{P_1}$ to gather information from $\boldsymbol{K_1}$ and $\boldsymbol{V_1}$, thereby presenting it to $\boldsymbol{Q_1}$ with reduced computational cost

$$\boldsymbol{P_1} = \text{agent_pool}\,(\boldsymbol{Q_1}) \in R^{n\times n\times C\times L_P} \tag{11}$$

where $\boldsymbol{P_1}$ is obtained by average pooling and a learnable linear projection, resulting in an agent vector of dimension $n \times n \times C \times L_p$, and $L_p = HW/N^2$, representing the number of sequence blocks.

3) Agent vectors $\boldsymbol{P_1}$ are then used in matrix multiplication operations with $\boldsymbol{Q_1}$ and $\boldsymbol{K_1}$ to obtain energy matrices $\boldsymbol{H}$ and $\boldsymbol{W}$, which encapsulate key features from $\boldsymbol{Q_1}$ and $\boldsymbol{K_1}$

$$\boldsymbol{H} = (\boldsymbol{Q_1} \otimes \text{transform}\ (\boldsymbol{P}_1)) \in R^{N\times N\times n\times n\times L_H} \tag{12}$$
$$\boldsymbol{W} = (\text{transform}\,(\boldsymbol{K_1}) \otimes \boldsymbol{P}_1) \in R^{n\times n\times N\times N\times L_W} \tag{13}$$

where $L_H = L_W = HW/N^2$, representing the number of sequence blocks.

4) The key information $\boldsymbol{W}$ from $\boldsymbol{V_1}$ and $\boldsymbol{K_1}$ is used in matrix multiplication to obtain energy matrix $\boldsymbol{U}$, and then energy matrices $\boldsymbol{H}$ and $\boldsymbol{WU}$ are combined through matrix multiplication and added to a position bias to derive the features $\boldsymbol{M}$ extracted by the NSA module

$$\boldsymbol{U} = (\boldsymbol{V} \otimes \boldsymbol{W}) \in R^{n\times n\times C\times L_U} \tag{14}$$
$$\boldsymbol{M} = (\boldsymbol{H} \otimes \boldsymbol{U} \oplus \text{ Position bias}) \in R^{N\times N\times C\times L_M} \tag{15}$$

where position bias represents the positional offset attempted as the input tensor is divided into sequence blocks and $L_U = L_M = HW/N^2$, indicating the number of sequence blocks.

5) The features $\boldsymbol{M}$ are then processed through window combination, transformation, and normalization operations to produce X', which serves as the input tensor for the global spectrum agent (GSA) attention module. The tensor X' is split into three equal parts to form the new $\boldsymbol{Q}_2$, $\boldsymbol{K}_2$, and $\boldsymbol{V}_2$

$$\boldsymbol{Q_2} = \text{query}\,(X') \in R^{H\times W\times C} \tag{16}$$
$$\boldsymbol{K_2} = \text{key}\,(X') \in R^{H\times W\times C} \tag{17}$$
$$\boldsymbol{V_2} = \text{value}\,(X') \in R^{H\times W\times C} \tag{18}$$

where $H\times W\times C$ represents the dimensions of the tensor.

6) Similar to the NSA module, the GSA attention module uses a small set of agent vectors $\boldsymbol{P_2}$ for global spectrum information modeling to reduce computational complexity

$$\boldsymbol{P_2} = \text{agent_pool}\ (\boldsymbol{Q_2}) \in R^{H\times W\times c} \tag{19}$$

where $\boldsymbol{Q_2}$ is transformed into an agent vector $\boldsymbol{P_2}$ of dimension $H \times W \times c$ through an average pooling and a learnable projection along the channel dimension, c represents the use of a lower-dimensional spectral channel number to characterize the original spectral channel number.

7) Agent vectors $\boldsymbol{P_2}$ are again used in matrix multiplication operations with $\boldsymbol{Q_2}$ and $\boldsymbol{K_2}$ to obtain new energy matrices $\boldsymbol{H'}$ and $\boldsymbol{W'}$, capturing key features from $\boldsymbol{Q_2}$ and $\boldsymbol{K_2}$

$$\boldsymbol{H'} = (\boldsymbol{Q_2} \otimes \text{ transform }\ (\boldsymbol{P_2})) \in R^{c\times C} \tag{20}$$
$$\boldsymbol{W'} = (\text{transform }\ (\boldsymbol{K_2}) \otimes \boldsymbol{P_2}) \in R^{C\times c} \tag{21}$$

where $L_H = L_W = HW/N^2$, representing the number of sequence blocks.

8) The key information $\boldsymbol{W'}$ from $\boldsymbol{V_2}$ and $\boldsymbol{K_2}$ is utilized in matrix multiplication to obtain energy matrix $\boldsymbol{U'}$, and then energy matrices $\boldsymbol{H'}$ and $\boldsymbol{U'}$ are combined through matrix multiplication to derive the features X_2 extracted by the GSA attention module

$$\boldsymbol{U'} = \left(\boldsymbol{V_2} \otimes \boldsymbol{W'}\right) \in R^{H\times W\times c} \tag{22}$$
$$X_2 = \text{transform}\left(\boldsymbol{H'} \otimes \boldsymbol{U'}\right) \in R^{H\times W\times C}. \tag{23}$$

C. Triplet MLP Module

To gain a deeper understanding of the inherent correlations within hyperspectral imagery, we analyze the intermediate hyperspectral feature tensor from a multimode unfolding perspective. As shown in Fig. 4, all three modes exhibit evident low-rank tendencies, as their singular values decay rapidly. This indicates that the hyperspectral feature tensor

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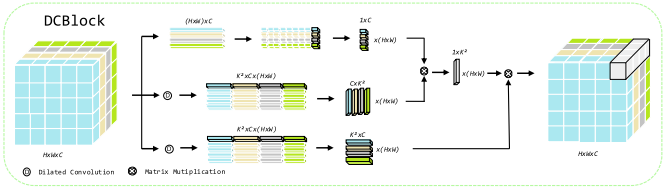


Fig. 3. Multiscale dilated convolution module is primarily used to aggregate spatial information at various scales in order to efficiently capture subtle local spatial and contextual patterns and provide multiscale spatial cues for subsequent attention modules.

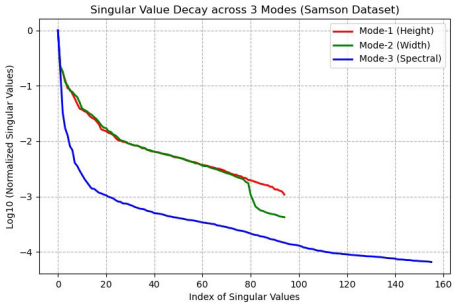


Fig. 4. Singular value decay analysis of intermediate features under three unfolding modes on the Samson dataset. The intermediate feature tensor is unfolded along the height, width, and spectral modes, respectively, and singular value decomposition is applied to each unfolded matrix. All three curves exhibit clear low-rank tendencies, while their distinct decay behaviors indicate that different unfolding modes capture complementary correlation structures.

contains structured correlations under all three unfolding views. More importantly, the decay curves are not identical across modes. The height-mode and width-mode curves are close but still show noticeable differences, suggesting that they capture similar yet anisotropic spatial structures. In contrast, the spectral-mode curve presents a distinctly different decay pattern, revealing another complementary correlation structure associated with spectral redundancy and endmember mixing. Therefore, it motivates us to design a Triplet MLP that separately learns height-wise, width-wise, and spectral-wise correlations and fuses them into a unified representation for hyperspectral unmixing.

The proposed Triplet MLP module in this method is distinct from traditional MLP models in that it not only preserves the original spatial dimensions but also encodes features along both height and width dimensions. This enables more effective capture of pixel-level details, thereby enhancing spatial resolution. In addition, traditional MLP models may lose positional information when processing 2-D image data. In contrast, the tripartite MLP, with its unique structure, is better able to preserve and utilize this information, achieving more accurate identification and localization of different substances during the unmixing process. By encoding features across different dimensions and aggregating them, the tripartite MLP module can produce complementary feature representations. This multidimensional feature fusion enhances the model's ability to recognize target substances, thus improving unmixing accuracy.

The execution process of the Triplet MLP Module is illustrated in Fig. 5, with the specific computation process as follows.

1) Assuming the tensor obtained through the S^2AM is $X_2 \in R^{B\times H\times W\times C}$, in the first branch, the input tensor X is divided into M parts out of N along the channel dimension, expanded along the height dimension to obtain tensor $\boldsymbol{H_1}$, which is then added to its result after passing through the height MLP, and adjusted in dimension to obtain $\boldsymbol{H_1}'$. In the second branch, X_2 is similarly divided into M parts out of N along the channel dimension, but it is expanded along the width dimension to obtain tensor $\boldsymbol{W_1}$, which is then added to its result after passing through the width MLP and adjusted in dimension to obtain $\boldsymbol{W_1}'$. The third branch sums the tensor X_2 directly with the result after passing through the channel MLP and adjusts the dimension to obtain C'. The computation process is as follows:

$$\boldsymbol{H_1} = H_\text{aggregation}\,(X_2) \in R^{B\times N\times W\times HM} \tag{24}$$

$$\boldsymbol{H_1}' = \text{transform}\,(H_mlp\,(\boldsymbol{H_1}) + \boldsymbol{H_1}) \in \mathbf{R}^{B\times H\times W\times C} \tag{25}$$

$$\boldsymbol{W_1} = H_\text{aggregation}\,(X_2) \in R^{B\times N\times M\times HW} \tag{26}$$

$$\boldsymbol{W_1}' = \text{transform}\,(W_mlp\,(\boldsymbol{W_1}) + \boldsymbol{W_1}) \in \mathbf{R}^{B\times H\times W\times C} \tag{27}$$

$$\boldsymbol{C}' = \text{transform}\,(C_mlp\,(X_2) + X_2) \in \mathbf{R}^{B\times H\times W\times C} \tag{28}$$

where $N = C/M$ and $HM = H \times M$ and functions $H_mlp()$, $W_mlp()$, and $C_mlp()$ perform MLP calculations along the height, width, and channel dimensions, respectively.

2) Finally, the outputs of these three branches are summed with the original tensor X_2, and then passed through a fully connected layer to obtain the final output $\boldsymbol{X_3}$ of the tripartite MLP, as shown in the following equation:

$$\boldsymbol{X_3} = FC\left(\boldsymbol{H_1}' + \boldsymbol{W_1}' + \boldsymbol{C}' + X_2\right) \in \mathbf{R}^{B\times H\times W\times C} \tag{29}$$

where FC represents the fully connected layer.

D. Decoding Module

The decoding module of this method includes a fully connected layer and a decoder, where the decoder is composed of two convolutional layers and activation functions. After passing through the first convolutional layer, an abundance matrix is output by the softmax function, and endmember spectra are obtained after the second convolutional layer. Specifically, the output from the triplet MLP module, denoted as $\boldsymbol{X_3} \in R^{B\times H\times W\times C}$, is processed through a fully connected layer to result in $\boldsymbol{X_3}' \in R^{P\times D}$, where P is the number of endmembers and $D = (H \times W)^{1/2}$. However, for unmixing purposes, $\boldsymbol{X_3}' \in R^{P\times D}$ is upsampled to a size of $P \times (H\cdot W)$. Upscaling from a relatively smaller dimension to $H \cdot W$ dimensions introduces noise in the final output. To mitigate this issue, a convolution operation featuring a kernel size of 3×3, a stride of 1, and padding of 1 is utilized. Subsequently, a reshaping operation transforms the output matrix into an

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Algorithm 1 Multiscale Spatial-Spectrum Agent Attention Network

Input: Hyperspectral data $X_0 \in R^{B\times H\times W\times C}$.

Initialization:

Epoch set to 200; batch size set to $H \times W$; patch size set to 5; learning rate set to $6e$-3; hyperparameter β set to 5000; α set to $1e-2$;

While Loop until the stopping condition is fulfilled:

1: Execute three parallel S^2AM to aggregate multi-scale spatial information, yielding an attention output;
2: Execute the spatial-spectrum agent attention module to efficiently aggregate and integrate local and global spatial-spectrum information;
3: Execute the triadic MLP module to continue encoding features across multiple dimensions;
4: Execute the decoding module, reshaping the results from Step into the abundance map $M_{out} \in R^{P\times H\times W}$ and adjust through backpropagation to refine the endmember spectra $E \in R^{Band\times P}$;

Output: Predict endmembers and abundance map.

abundance map $M_{out} \in R^{P\times H\times W}$. The abundance matrix's values are required to adhere to the ANC and ASC constraints. For the calculation of endmembers, the abundance matrix M_{out} undergoes processing via a second convolutional layer. This convolutional operation expands the band count in M_{out} from P to Band, facilitating the decoding of the HSI. Convolutional layer weights, initially set with endmembers derived from VCA, undergo adjustments through the backpropagation of gradients during the learning phase, aiming to estimate the final endmembers $E \in \mathbf{R}^{Band\times P}$.

E. Implementation

We show the steps of the network structure in Algorithm 1 to understand.

III. Experimental Section

A. Dataset Description

The M600 PRO six-rotor UAV from Shenzhen DJI Innovation Company served as the flight platform for the acquisition of the first two crop unmixing datasets. The hyperspectral imager utilized the GaiaSky-mini push-broom airborne hyperspectral imaging system developed by Sichuan Shuangli Spectrum Company. The spectral range of the system was 400–1000 nm, with a resolution of 3.5 ± 0.5 nm and 177 bands. The system was equipped with a 23-mm imaging lens, which operated in hover mode for reflection acquisition. To minimize the influence of the sun's altitude angle variations on HSI acquisition, automatic exposure calibration was conducted using a whiteboard during each takeoff, followed by reflectance calculation. In addition, after each flight, the UAV collected HSIs of a gray cloth (with a reflectivity of 50%) for subsequent atmospheric correction. The gray cloth was positioned at the same altitude as the correction data to ensure accurate reflectance measurements. In the two crop datasets produced, all HSIs captured by the UAV were processed using the Spectra-VIEW software, which provided lens

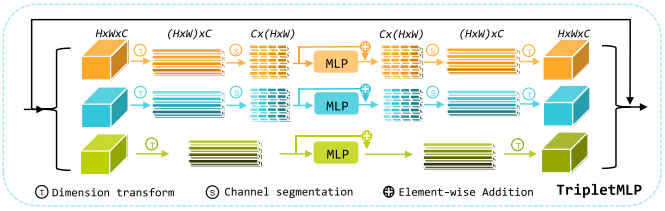


Fig. 5. Triplet MLP module mainly improves the model's capability to identify target substances by processing feature representations across the dimensions of height, width, and space.

correction, uniformity adjustment, and reflectance calibration, resulting in the final hyperspectral reflectance images suitable for decontamination. The images were acquired at an altitude of 300 m with a spatial resolution of 0.63 m. Concurrently, a high-resolution UAV-visible light camera (operating at an altitude of 50 m with a spatial resolution of approximately 7 cm) was employed to capture visible light images of the same study area. Both image types contained latitude and longitude information, facilitating precise alignment of images from different altitudes or sensors via georeferencing. The crop HSIs and the high-resolution UAV orthophoto images were registered using pixel fusion. This approach allowed for the examination of corresponding high-resolution orthophoto pixels within the coverage area of a hyperspectral pixel. If these pixels corresponded to the same crop type, the hyperspectral pixel was marked as a pure pixel [37], [45]. Abundance labels were manually annotated based on visual interpretation of the imagery.

Experiments were conducted on four distinct datasets, described as follows.

1) *ChiTian Crop Dataset:* The dataset under consideration is derived from the crop research area located around Chitian Reservoir in Hainan Province, with imaging performed on March 16, 2023. This dataset includes a proprietary crop hyperspectral decomposition dataset consisting of 96 × 96 hyperspectral pixels across 177 spectral bands, spanning the 400–1000-nm wavelength range. In addition, four primary pure endmember types—soil, rubber trees, coconut trees, and mango trees—were extracted from the imagery. The specific abundance distribution of these endmembers was determined through manual labeling using high-resolution visible light images, as shown in Fig. 6(a).
This dataset exhibits complex background interference mainly because multiple vegetation types coexist with bare soil and field roads within small spatial regions. Their spectral signatures partially overlap in the visible-near-infrared range, generating mixed pixels particularly along crop boundaries and field edges. These characteristics make the dataset suitable for evaluating unmixing algorithms under realistic spatial–spectral complexity.
2) *WenChange Crop Dataset:* The Wenchang crop hyperspectral unmixing dataset was compiled by students in the laboratory from the HSI data collection project in Wenchang, Hainan Province. The dataset [see Fig. 6(b)] contains 120 × 120 hyperspectral pixels, covering a

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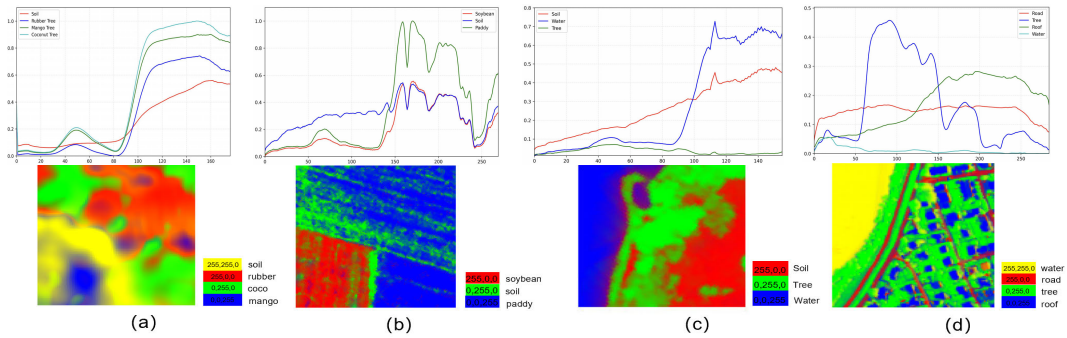


Fig. 6. GT endmember spectra and abundance chart on four datasets. (a) ChiTian Crop Dataset. (b) WenChang Crop Dataset. (c) Samson Dataset. (d) Apex Dataset.

spectral range from 400 to 1000 nm. Following the endmember extraction principle, three primary pure endmembers were identified and extracted from the dataset. The specific abundance distributions of these endmembers were obtained through manual visual interpretation, aided by high-resolution visible light imagery.

Compared with the ChiTian dataset, the WenChang dataset presents a different form of complexity: crop–soil mixtures occur in narrow planting strips, and soil moisture variation produces subtle spectral gradients, increasing the difficulty of distinguishing soybean from soil and paddy. These conditions serve as another representative type of complex background for evaluating spatial–spectral unmixing robustness.

3) *Samson Dataset [46]:* The Samson dataset, widely recognized for hyperspectral unmixing applications, comprises 95 × 95 hyperspectral pixels distributed over 15 distinct spectral bands, covering wavelengths from 401 to 889 nm. Within this dataset, three primary endmembers were identified. The authentic spectral signatures of these endmembers, as depicted in Fig. 6(c), were discerningly chosen directly from the dataset.
4) *Apex Dataset [22]:* The Apex dataset, publicly available for unmixing tasks, contains 110 × 110 hyperspectral pixels across 285 spectral bands, spanning wavelengths from 413 to 2420 nm. Within this dataset, four distinct endmembers are present. The spectral signatures of these endmembers, depicted in Fig. 6(d), were derived directly from the dataset.

   This dataset represents a more challenging urban form of complex background interference: artificial materials (roads and roofs) possess reflectance characteristics partially similar to vegetation in several spectral intervals, while water introduces strong low-reflectance regions. In addition, the high-frequency spatial structures of man-made objects create mixed pixels at the submeter scale, producing substantial spectral–spatial ambiguity. This makes APEX an ideal benchmark for evaluating robustness under cross-class spectral similarity.

It is worth noting that ChiTian and WenChang are crop-specific UAV datasets designed for evaluating unmixing performance in realistic agricultural scenarios, whereas Samson and APEX are widely used public benchmarks that contain various noncrop endmembers (e.g., roads, roofs, and water). We therefore use the former to validate the effectiveness of MS^{2}ANet in complex crop scenes, and the latter to assess its generalization ability and to enable fair comparison with existing unmixing methods.

Across all four datasets, the forms of complex background interference differ, ranging from crop–soil–road mixtures and tangled agricultural boundaries to urban materials with partially overlapping spectra, which allows a comprehensive evaluation of the proposed model under multiple spatial–spectral complexity conditions.

B. Experimental Setup

1) Evaluation Metrics: For training the proposed model, a combination of two distinct loss functions was utilized: the mean square error (mse) objective function and the SAD loss [47]. The root mse (RMSE) function was employed to calculate the loss between estimated abundance scores and GT abundance scores, while the SAD loss function was used to measure the loss between estimated endmembers and the true endmembers

$$Y_{re}\left(X, X'\right) = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left(X'_{\mathrm{ij}} - X_{\mathrm{ij}}\right)^2 \tag{30}$$

$$Y_{\mathrm{sad}}\ \left(X, X'\right) = \frac{1}{P} \sum_{i=1}^{P} \arccos\left(\frac{\left(X_i, X'_i\right)}{\|X_i\|_2 \left\|X'_i\right\|_2}\right) \tag{31}$$

where X and X' represent the ground truth (GT) and the estimated reconstructed images (or abundance maps), respectively. H and W denote the spatial height and width of the image. X_{ij} and X'_{ij} denote the pixel values at the spatial location (i, j). In addition, P represents the total number of endmembers.

The RE loss, calculated using the mse objective function, aids in acquainting the model with the fundamental characteristics of the input HSI. Being a scale-invariant objective function, the inclusion of the SAD loss promotes quicker convergence of the model as a whole. The total loss is determined by the weighted sum of these two losses

$$Y = \beta Y_{\mathrm{re}}\ + \gamma Y_{\mathrm{sad}} \tag{32}$$

where β and γ are hyperparameters.

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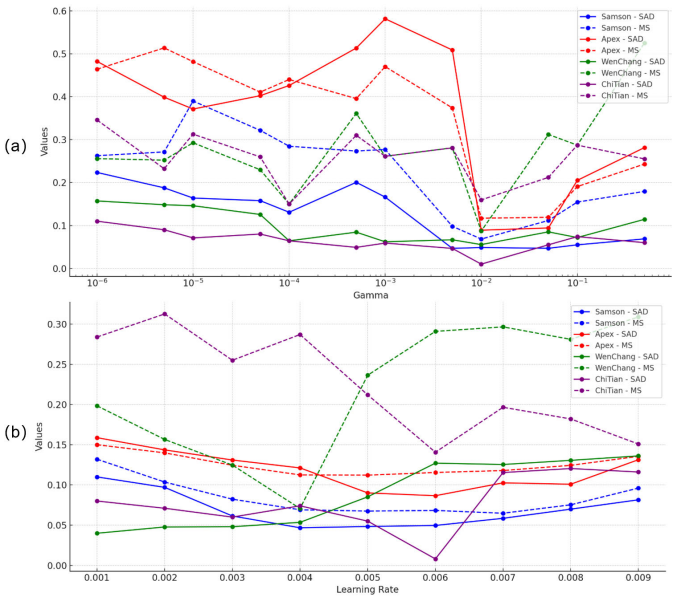


Fig. 7. (a) Results of varying gamma values from $1e^{-6}$ to $1e^{-1}$ at ten-fold intervals, with the supervariable β set to 5000 for all four datasets, to obtain the corresponding SAD and RMSE values. (b) Optimal result from (a), the learning rate is further adjusted to achieve the best SAD and RMSE values.

TABLE I

RESULTS OF THE CHITIAN CROP DATASET ON DIFFERENT NETWORKS USING THE SAD EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

ChiTian	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Coco	0.0954	0.0312	0.0305	0.0285	0.0215	0.0143	0.0185	0.0126	**0.0100**
Rubber	0.1317	0.0266	0.0246	0.0210	0.0187	0.0186	0.0118	0.0123	**0.0102**
Mango	0.0746	0.0472	0.0395	0.0212	0.0139	0.0132	0.0144	0.0105	**0.0057**
Soil	0.0320	0.0450	0.0366	0.0212	0.0161	0.0161	0.0098	0.0122	**0.0063**
Overall	0.0847	0.0375	0.0328	0.0230	0.0176	0.0156	0.0136	0.0119	**0.0080**

TABLE II

RESULTS OF THE CHITIAN CROP DATASET ON DIFFERENT NETWORKS USING THE RMSE EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

ChiTian	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Coco	0.2874	0.3514	0.3261	0.3400	0.3315	0.2972	0.2784	0.1534	**0.0839**
Rubber	0.3305	0.3189	0.3308	0.3317	0.3264	0.3091	0.2075	0.2314	**0.1983**
Mango	0.2036	0.2558	0.1901	0.1977	0.1848	0.1883	0.1994	0.1904	**0.1835**
Soil	0.2876	0.2340	0.1505	0.1471	0.0981	0.1354	0.1088	0.1132	**0.0974**
Overall	0.2998	0.2938	0.2620	0.2675	0.2550	0.2437	0.1985	0.1721	**0.1407**

2) Experimental Configuration: The model was developed using PyTorch and tested on a workstation equipped with an NVIDIA Quadro RTX 5000 16 GB GPU. For MS²ANet, the Adam optimizer was used with a learning rate of $6e^{-3}$, a hyperparameter β set to 5000, and γ set to $1e^{-2}$. Similarly, for MS²ANet, the Adam optimizer was applied, with β fixed at 5000 for consistent control of variables. The optimal settings for γ and the learning rate are shown in Fig. 7.

C. Experimental Results

The performance of the proposed model was evaluated and compared with eight different unmixing techniques from various categories: VCA+FCLS [4], HyperspecAE [46], CyCUNet [48], MiSiCNet [31], HSU [22], IDNet [49], SWCNet [50], and SSAFNet [51].

1) *ChiTian Crop Dataset:* Quantitative results for the ChiTian Crop Dataset are presented in Tables I and II, with visual comparisons shown in Figs. 8 and 9. In terms of abundance and endmember accuracy, the proposed model exhibits significantly superior unmixing performance compared to baseline methods: the average SAD is 0.0080, and the average RMSE is 0.1707. Visual analysis indicates that baseline algorithms struggle to distinguish between rubber trees and coconut trees: for instance, the bottom-right corner of the abundance map (which should correspond to coconut trees) is misclassified as rubber trees, and the two classes are frequently confused at parcel boundaries. In contrast, MS²ANet produces sharper abundance transitions and fewer misclassified boundary pixels. This behavior stems from baseline methods' inability to simultaneously model multiscale spatial context and long-range spectral dependencies, leading to misinterpretation of mixed pixels at crop boundaries. By contrast, MS²ANet's multiscale module provides fine-to-coarse spatial cues, while its spatial-spectrum agent attention enhances differentiation in mixed regions, enabling accurate separation of visually similar vegetation types. This aligns with our design logic: multiscale dilated convolutions aggregate local boundary context, and the nonlocal spatial agent module captures long-range dependencies between homogeneous crop parcels, ensuring consistent abundance estimates within individual fields.
Furthermore, the endmember curves reconstructed by MS²ANet are more closely aligned with the GT, particularly at key absorption troughs and reflectance peaks. This demonstrates effective suppression of spectral noise and recovery of class-specific spectral characteristics, further validating the model's robustness in complex crop environments. For mango tree and soil abundance extraction, all baseline algorithms yield reasonable estimates; however, the proposed method is closest to the GT, especially for challenging classes like rubber trees and coconut trees.

TABLE III

RESULTS OF THE WENCHANG CROP DATASET ON DIFFERENT NETWORKS USING THE SAD EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

WenChang	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Soybean	0.1604	0.1793	0.1433	0.1658	0.1051	0.1692	0.1237	0.1002	**0.0912**
Soil	0.1418	0.0954	0.0716	0.0740	0.0636	0.0594	0.0612	0.0587	**0.0511**
Paddy	0.2643	0.0361	0.0245	0.0215	0.0156	**0.0132**	0.0196	0.0188	0.0184
Overall	0.2176	0.1369	0.0798	0.0871	0.0614	0.0806	0.0673	0.0592	**0.0505**

TABLE IV

RESULTS OF THE WENCHANG CROP DATASET ON DIFFERENT NETWORKS USING THE RMSE EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

WenChang	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Soybean	0.3593	0.1839	0.1312	0.0651	0.0673	0.0664	0.0647	0.0632	**0.0596**
Soil	0.1376	0.1573	0.1120	**0.0619**	0.0632	0.0645	0.0653	0.0636	0.0647
Paddy	0.3427	0.1798	0.1545	0.0895	0.0843	0.0857	0.0880	**0.0860**	0.0878
Overall	0.2654	0.1736	0.1325	0.0722	0.0716	0.0722	0.0727	0.0709	**0.0707**

2) *WenChang Crop Dataset:* Quantitative results for the Samson dataset are documented in Tables III and IV, complemented by visual comparisons in Figs. 10 and 11. In terms of abundance and endmember accuracy, the proposed MS²ANet outperforms

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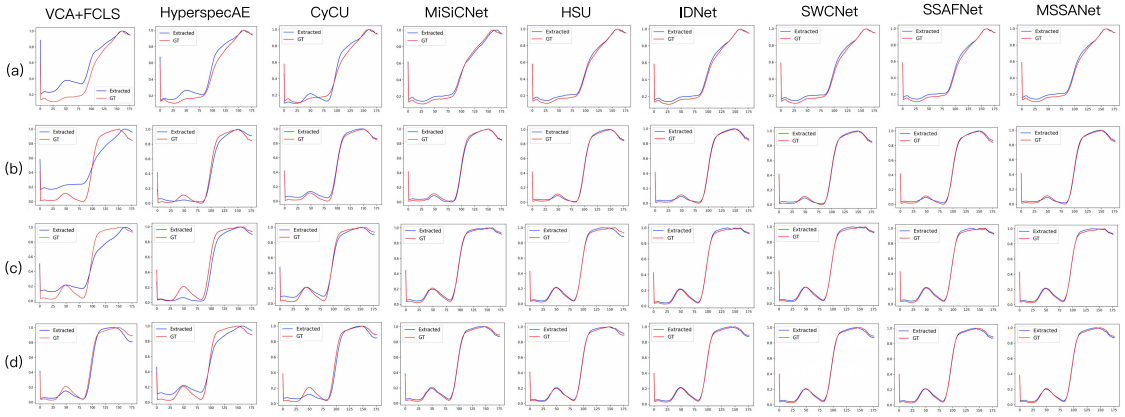


Fig. 8. Endmember unmixing effect images obtained from the ChiTian Crop Dataset on different networks. (a) Rubber. (b) Coco. (c) Mango. (d) Soil.

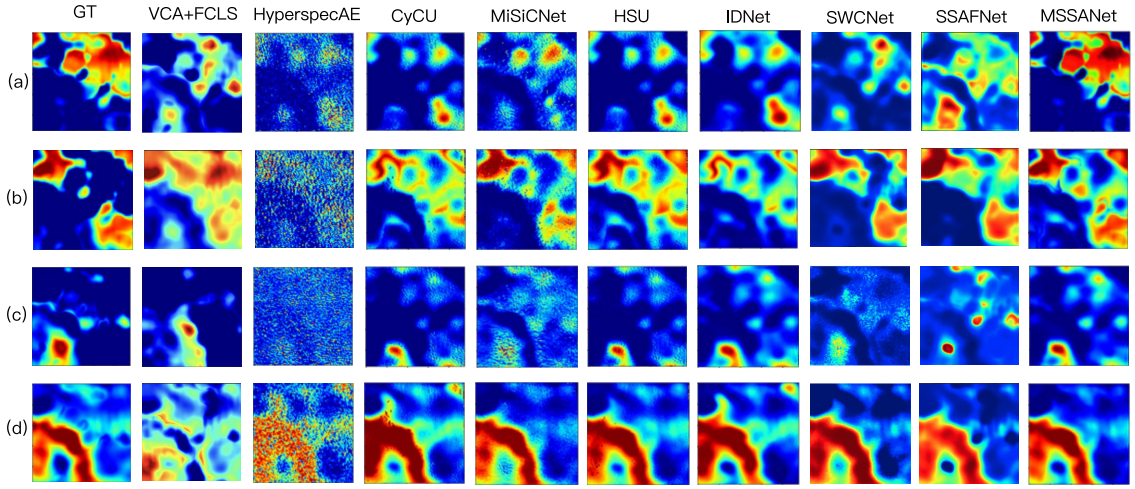


Fig. 9. Abundance map effect images and endmember unmixing effect images obtained from the ChiTian Crop Dataset on different networks. (a) Rubber. (b) Coco. (c) Mango. (d) Soil.

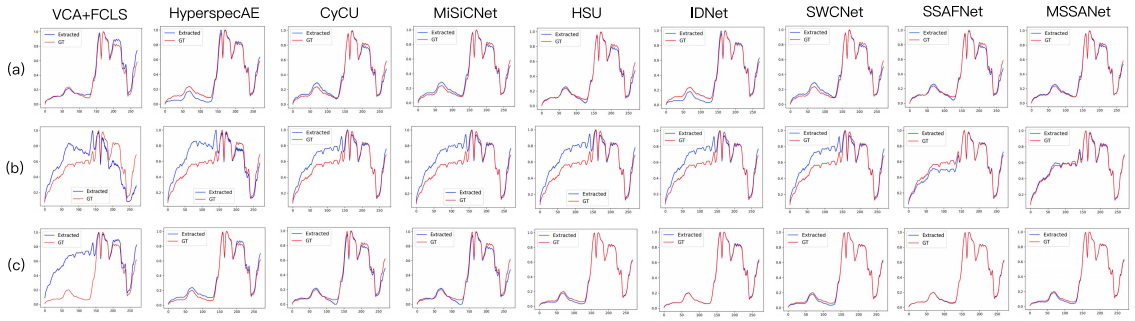


Fig. 10. Endmember unmixing effect images and endmember unmixing effect images obtained from the WenChang Dataset on different networks. (a) Soybean. (b) Soil. (c) Paddy.

competing methods, particularly in soybean and soil extraction—with an average SAD of 0.0505 and RMSE of 0.0707. Analysis of the abundance maps reveals deeper insights. While all six algorithms effectively distinguish spatial characteristics of land-cover types, IDNet and MS^{2}ANet show high similarity to GT abundance—with MS^{2}ANet excelling in subtle soybean and soil extraction. Specifically, soybean and soil exhibit gradual spectral transitions due to field texture and moisture variations, posing challenges for baseline methods: most baselines over-smooth these gradients, yielding patchy or overly homogeneous abundance maps. By contrast, MS^{2}ANet generates smoother yet structurally consistent abundance distributions, capturing local spatial continuity without sacrificing material-specific spectral discrimination.

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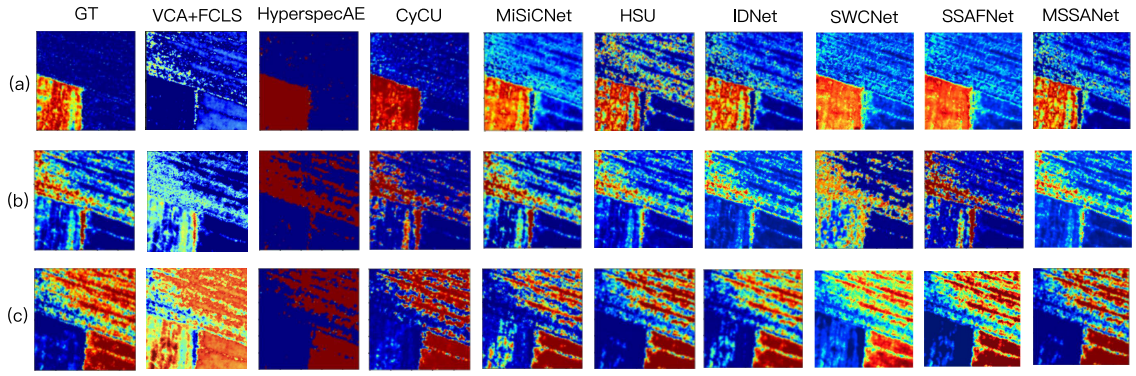


Fig. 11. Abundance map effect images obtained from the WenChang Dataset on different networks. (a) Soybean. (b) Soil. (c) Paddy.

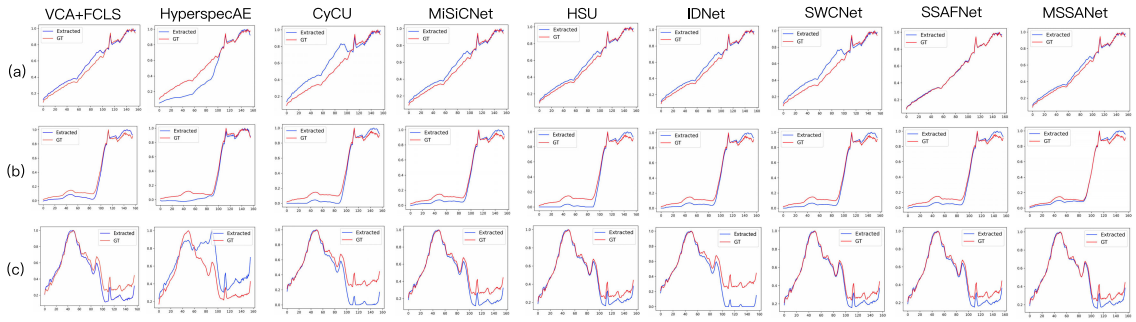


Fig. 12. Endmember unmixing effect images and endmember unmixing effect images obtained from the Samson Dataset on different networks. (a) Soil. (b) Tree. (c) Water.

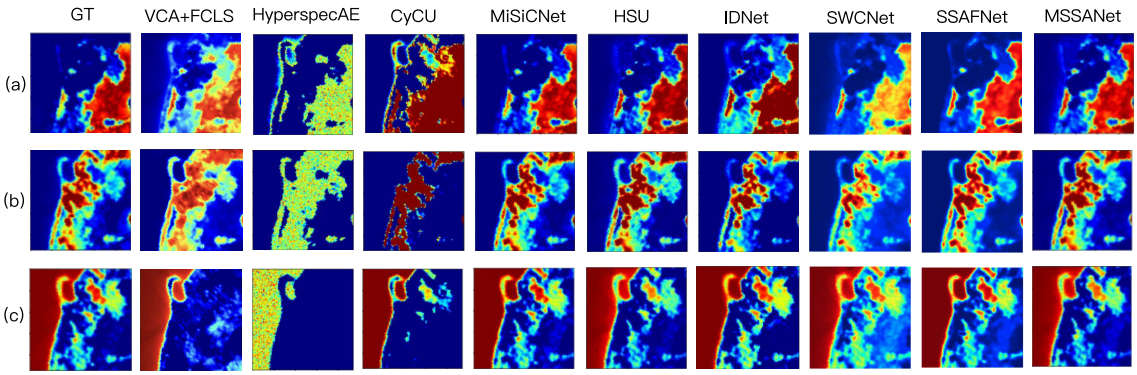


Fig. 13. Abundance map effect images obtained from the Samson Dataset on different networks. (a) Soil. (b) Tree. (c) Water.

The endmember spectra reconstructed by MS^{2}ANet also retain clearer low-frequency structures and accurate reflectance levels, indicating that the spatial-spectrum agent mechanism effectively regularizes spectral variation during unmixing. Furthermore, MS^{2}ANet aligns more closely with GT in endmember spectra, confirming superior endmember reconstruction performance.

3) *Samson Dataset:* Quantitative outcomes for the Samson dataset are documented in Tables V and VI, complemented by visual comparisons in Figs. 12 and 13. Identifying the water endmember posed a considerable challenge for alternative approaches: specifically, the dataset features sharp spectral transitions at water–soil boundaries, and most comparison models fail to reconstruct water endmembers accurately as they cannot model global spectral correlations, leading to mixing with dark soil spectra. By contrast, MS^{2}ANet successfully isolates water signatures via its GSA. And it enhances discrimination of long-range spectral dependencies, thereby delivering the most reliable water endmember estimation.

TABLE V

RESULTS OF THE SAMSON DATASET ON DIFFERENT NETWORKS USING THE SAD EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Samson	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Soil	0.1525	0.8916	0.2071	0.1500	0.1019	0.2547	0.1299	**0.0607**	0.0637
Tree	0.0316	0.0672	0.0378	0.0334	0.0518	0.0297	0.0300	0.0302	**0.0266**
Water	0.0753	0.8491	0.1976	0.0860	0.0952	0.0574	0.0803	0.0553	**0.0547**
Overall	0.0935	0.7360	0.1475	0.0898	0.0830	0.1139	0.0800	0.0487	**0.0483**

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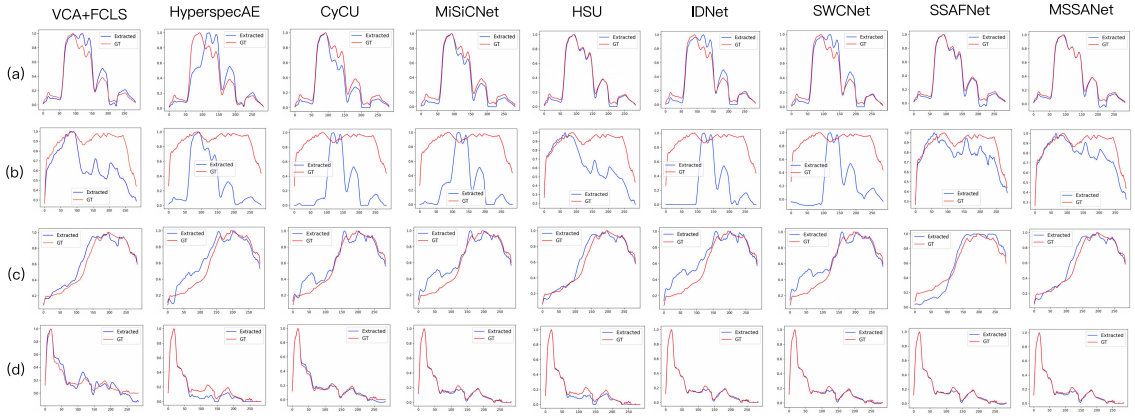


Fig. 14. Endmember unmixing effect images and endmember unmixing effect images obtained from the APEX Dataset on different networks. (a) Road. (b) Tree. (c) Roof. (d) Water.

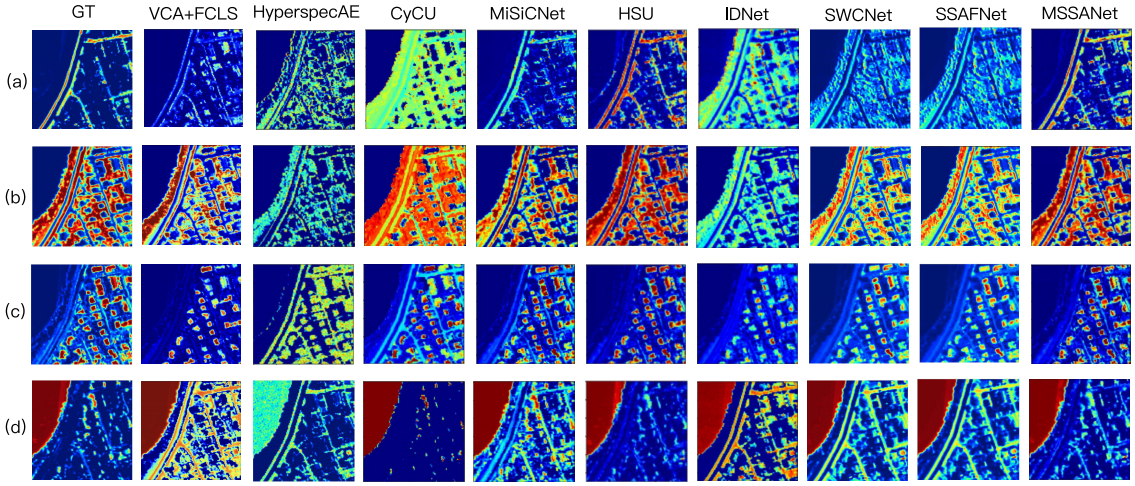


Fig. 15. Abundance map effect images obtained from the APEX Dataset on different networks. (a) Road. (b) Tree. (c) Roof. (d) Water.

TABLE VI

RESULTS OF THE SAMSON DATASET ON DIFFERENT NETWORKS USING THE RMSE EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Samson	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Soil	0.2103	0.3742	0.1897	**0.0619**	0.0632	0.0784	0.0842	0.0684	0.0646
Tree	0.3285	0.4140	0.1735	**0.0651**	0.0673	0.0726	0.0910	0.0670	0.0667
Water	0.3567	0.3961	0.1743	0.0895	0.0843	0.0527	0.0698	**0.0522**	0.0711
Overall	0.2879	0.3948	0.1792	0.0722	0.0716	0.0679	0.0816	**0.0625**	0.0675

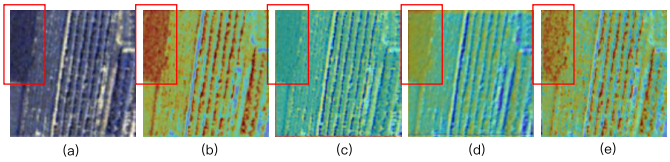


Fig. 16. Heatmap visualization of different spatial attention on crop scenery. (a) Original map. (b) Proposed NSA. (c) EMA [52]. (d) CBAM [53]. (e) NonLocal Block [54].

In terms of SAD, MS^{2}ANet achieved the best average score of 0.0483. Regarding the RMSE evaluation, although our proposed model did not achieve the best score in any of the three individual categories (soil, tree, and water) on the Samson dataset, it attained the second-best overall RMSE of 0.0675, closely trailing SSAFNet's 0.0625. This indicates that while some competing methods may excel in specific endmembers but fluctuate in others, MS^{2}ANet maintains a highly balanced and stable performance across all categories, avoiding severe performance degradation in any single class. This validates the overall robustness and adaptability of the proposed method in complex scenarios.

TABLE VII

RESULTS OF THE APEX DATASET ON DIFFERENT NETWORKS USING THE SAD EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Apex	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Road	0.3762	0.4261	0.3794	0.1879	**0.0923**	0.3832	0.3100	0.0961	0.0942
Tree	0.1417	0.3415	0.1827	0.1764	0.1281	0.1896	0.1698	0.1266	**0.1224**
Roof	0.1017	0.2587	0.1167	0.1198	0.0922	0.1376	0.1355	0.1025	**0.0913**
Water	0.2475	0.2114	0.0384	0.0354	0.0429	**0.0287**	0.0342	0.0356	0.0384
Overall	0.2766	0.3094	0.1793	0.1299	0.0889	0.1848	0.1626	0.0902	**0.0866**

4) *APEX Dataset:* Quantitative results for the APEX dataset can be found in Tables VII and VIII, with visual comparisons in Figs. 14 and 15. Unmixing results for the APEX dataset were not particularly good across many models, making it a challenging dataset to test the

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TABLE VIII

RESULTS OF THE APEX DATASET ON DIFFERENT NETWORKS USING THE RMSE EVALUATION METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Apex	VCA+FCLS	HyperspecAE	CyCU	MiSiCNet	HSU	IDNet	SWCNet	SSAFNet	Proposed
Road	0.4201	0.3249	0.8731	0.1978	0.1744	0.4182	0.2685	0.2893	**0.1708**
Tree	0.1765	0.2693	0.7412	0.1563	0.1021	0.3123	0.1856	0.1820	**0.0818**
Roof	0.1377	0.3217	0.2465	**0.1114**	0.1211	0.1871	0.1420	0.1452	0.1201
Water	0.3012	0.3642	0.1463	0.2147	0.0929	0.2976	0.1952	0.1680	**0.0893**
Overall	0.2796	0.3200	0.5018	0.1700	0.1267	0.3094	0.1978	0.1961	**0.1155**

efficacy of our proposed model. Unlike crop datasets, the APEX scene contains urban materials with complex high-frequency spatial structures. Models relying exclusively on CNNs or transformers face difficulties in jointly capturing these high-frequency edges and global spectral relations, leading to fragmented or overly smoothed abundance maps. Specifically, our proposed MS²ANet excels in these challenging regions due to two key design choices. The multiscale module preserves multiresolution spatial structures, retaining fine details like high-frequency urban structures. The proposed GSA attention mechanism accurately reconstructs the distinct spectral profiles of artificial materials. These advantages translate to more realistic material separations and cleaner abundance boundaries.

Quantitatively, our model achieves the optimal average unmixing performance with an RMSE of 0.1155 and an average SAD of 0.0866. In detail, the HyperspecAE and CyCU methods performed poorly in terms of RMSE and SAD values. Visually, only the HSU method and our proposed method produced abundance maps closely resembling the GT, with other methods lagging behind. From the perspective of endmember images, the unmixing results of HyperspecAE and CyCU were particularly poor, while the HSU method and our proposed method appeared slightly superior visually.

D. Attention Visualization

To intuitively demonstrate the spatial perception ability of the proposed NSA in the context of hyperspectral unmixing, we conducted a visual comparison between our method and other popular spatial attention mechanisms (CBAM [53], Non-Local Block [54], and EMA [52]). The visualized results are presented in Fig. 16. Through this comparison, we draw the following conclusions: methods based on nonlocal attention mechanisms, specifically (a) NSA and (d) NonLocal Block exhibit significantly stronger spatial perception in the cotton region than (b) EMA and (c) CBAM; furthermore, NSA outperforms NonLocal Block in identifying indistinct global boundaries when extracting the overall contour of the cotton region.

E. Ablation Experiments

1) To evaluate the importance and effectiveness of each module in the proposed MS²ANet model, ablation experiments were conducted using the WenChang Crop dataset. In this section, the MS²ANet model was compared with four ablation versions as follows.

1) The model without the multiscale attention (MSA) module, denoted as w/o MSA.
2) The model without the NSA module, recorded as w/o NSA.
3) The model without the GSA module, labeled as w/o GSA.
4) The model without the triplet-based MLP model, labeled as w/o TMLP.

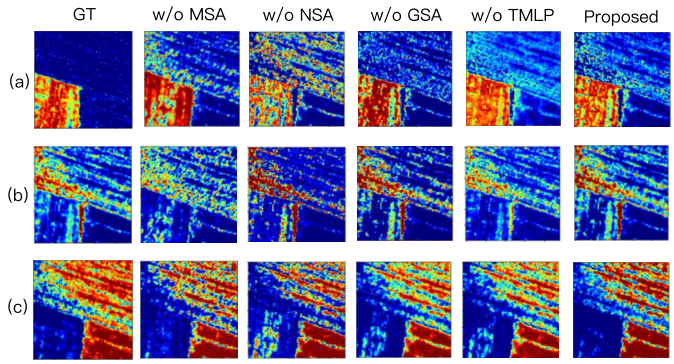


Fig. 17. Abundance map renderings of the ablation experiments on the Wenchang dataset. (a) Soybean. (b) Soil. (c) Paddy.

The quantitative results of the ablation experiment are presented in Table IX, and the qualitative results are shown in Figs. 17 and 18.

All ablation experiments are conducted on the WenChang crop dataset for consistency. As shown in Fig. 17, removing the MSA (w/o MSA) leads to blurred abundance transitions along field borders, indicating that the multiscale local context is crucial for boundary pixels. When the nonlocal spatial agent block is removed (w/o NSA), the abundance maps become spatially fragmented within large crop parcels, which confirms the importance of nonlocal spatial modeling. The w/o GSA variant fails to suppress highly correlated spectral components, resulting in less accurate endmember curves. Finally, without the triplet MLP (w/o TMLP), the model is less sensitive to subtle texture variations, and some small soybean patches are underestimated.

All ablation experiments are conducted on the WenChang crop dataset for consistency. These ablation observations also reveal how each module contributes to the overall unmixing process as follows.

1) w/o MSA loses fine spatial cues, causing abundance discontinuities in narrow boundary areas.
2) w/o NSA lacks global spatial context, resulting in misclassification of homogeneous regions with similar spectral patterns.
3) w/o GSA cannot accurately reconstruct endmembers requiring long-range spectral modeling, leading to distorted spectral curves.
4) w/o TMLP weakens spatial–spectral fusion, reducing its ability to recover subtle spectral variations in mixed pixels.

Although the abundance maps of ablation variants may not show significant differences at the global visual scale, the improvements introduced by the full MS²ANet are mainly reflected in local high-ambiguity regions, such as crop–soil transition zones, mixed boundaries, and regions with subtle spectral variations. These improvements, which include

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TABLE IX

QUANTITATIVE EXPERIMENTAL RESULTS OF THE MS^{2}ANET MODEL AND FOUR ABLATION MODELS

Metric	WenChang Crop Dataset				
	w/o MSA	w/o NSA	w/o GSA	w/o TMLP	Ours
SAD	0.0651	0.0647	0.0887	0.0535	**0.0505**
RMSE	0.0816	0.0803	0.0776	0.0748	**0.0707**

TABLE X

RESULTS OF ABLATION EXPERIMENTS ON AGENT MECHANISMS OF NSA AND GSA ON THE WENCHANG DATASET

Metric	WenChang Crop Dataset			
	w/o both Agent	w/o NSA Agent	w/o GSA Agent	Ours
SAD	0.0512	0.0532	0.0516	**0.0505**
RMSE	0.0713	0.0733	0.0719	**0.0707**
Params	6,122,753	5,924,456	5,835,452	**5,635,935**
Flops	3,846,265,463	3,723,475,634	3,686,285,017	**3,475,458,625**

TABLE XI

RESULTS OF ABLATION EXPERIMENTS ON MULTISCALE DILATION CONVOLUTION OF MSA ON THE WENCHANG DATASET

Metric	WenChang Crop Dataset				
	w/o MDC	[0,1,2]	[0,2,4]	[0,1,2,3,4]	[0,1,2,3]
SAD	0.0634	0.0538	0.0561	0.0517	**0.0505**
RMSE	0.0809	0.0734	0.0758	0.0711	**0.0707**

sharper boundary delineation, reduced abundance bleeding, and more stable decomposition of mixed pixels, are not always fully perceivable in low-resolution visualizations. However, they lead to consistent and meaningful gains in SAD/RMSE metrics and are clearly observable when examining local patches.

2) To verify the effectiveness of the proposed agent mechanism in both nonlocal spatial attention and global spectral attention, we conducted ablation validation on the Wenchang Crop dataset. The ablation details include the following four scenarios.

1) Neither agent mechanism is used (w/o both agent).
2) Only the global spectral attention employs the agent mechanism (w/o NSA agent).
3) Only the nonlocal spatial attention employs the agent mechanism (w/o GSA agent).
4) Both attention modules use the agent mechanism (ours).

The experimental results are presented in Table X. We found that the combination with both agent mechanisms achieved the best unmixing performance. Our analysis indicates that the agent mechanism can reduce redundant spatial and spectral correlations in attention computation while preserving the global feature representation capability. It also avoids high complexity pairwise calculations across all pixels or all spectra and enhances the model's ability to capture boundary and mixed spectral structures. In addition, we show the model parameters and FLOPs count under different scenarios in Table X. Our proposed method also exhibits optimal efficiency.

3) To justify the choice of dilation rates in the proposed multiscale dilated convolution module, we conducted an ablation study evaluating several dilation configurations, including without multiscale dilated convolution (w/o MDC), [0, 1, 2], [0, 2, 4], [0, 1, 2, 3, 4], and our final setting [0, 1, 2, 3]. The experimental results are summarized in Table XI.

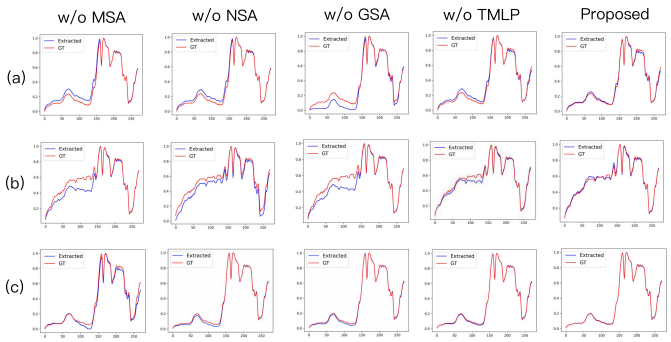


Fig. 18. Endmember unmixing effect images obtained from the WenChang Crop Dataset on different networks. (a) Soil. (b) Soybean. (c) Paddy.

We observe that the proposed dilation combination achieves the best unmixing performance. This can be attributed to the following reasons.

1) *Capturing Fine-Scale Boundary Texture:* The dilation rates 0 and 1 allow the model to capture detailed crop boundaries, edge gradients, and subtle spatial transitions that are critical for distinguishing mixed pixels along field edges.
2) *Modeling Mid-Scale Spatial Dependencies:* Dilation rates 2 and 3 enable the network to aggregate contextual information over larger field regions without excessive blurring, which helps resolve spectral mixing patterns caused by crop–soil transitions and irregular field geometry.
3) *Complementary Multiscale Representation:* Using all four dilation levels together provides a smooth and continuous receptive-field hierarchy, effectively balancing local detail extraction and global context modeling.

F. Spatial Visualization of Abundance Errors

Although RMSE and SAD quantitatively measure the overall unmixing quality, they are global averaged metrics and do not explicitly reveal where the prediction errors are spatially distributed. Since the proposed MS^{2}ANet is designed to improve the modeling of crop boundaries and mixed pixels, we further provide abundance error maps as a spatially explicit complement to the numerical evaluation.

For each dataset, we compute the absolute error between the predicted abundance map and the corresponding GT abundance map on a pixel-wise basis, and visualize the resulting residual distribution under a unified color scale. The corresponding results are shown in Fig. 19. It can be observed that the major errors of abundance estimation are not randomly distributed, but mainly appear near semantic transition zones, object contours, narrow boundary structures, and other highly mixed regions. This spatial pattern is consistent with the intrinsic difficulty of hyperspectral unmixing, where boundary pixels usually contain stronger material mixtures and more ambiguous spectral responses.

More importantly, the error maps show that MS^{2}ANet yields lower residual intensity and more compact error regions in these challenging locations. On the crop datasets, the advantage is particularly clear around crop–soil transitions and parcel boundaries, where the proposed method better

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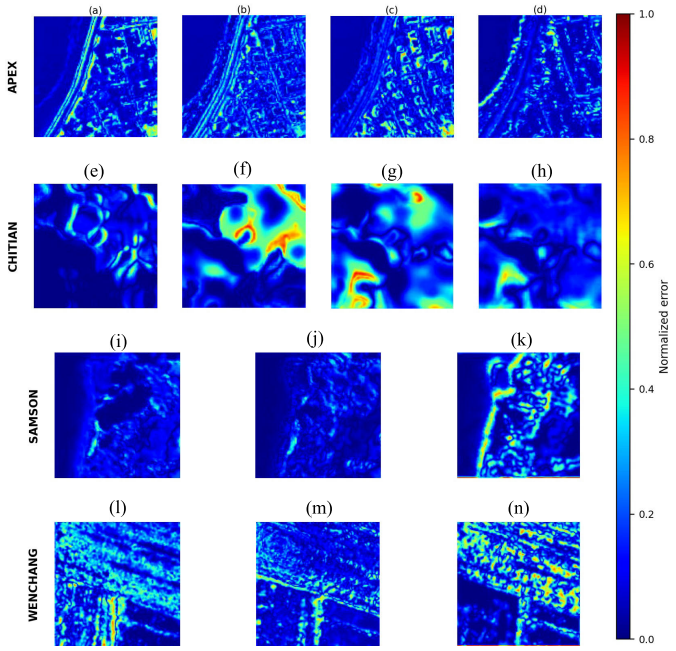


Fig. 19. Overview of abundance error maps of MS²ANet on the ChiTian, WenChang, Samson, and APEX datasets. The maps visualize the pixel-wise absolute difference between the estimated abundances and the GT under a unified color scale. The results show that the residuals are mainly concentrated near boundaries and mixed-pixel regions, while MS²ANet maintains relatively lower error intensity in these challenging areas. (a) Road. (b) Tree. (c) Roof. (d) Water. (e) Rubber. (f) Coco. (g) Mango. (h) Soil. (i) Soil. (j) Tree. (k) Water. (l) Soybean. (m) Soil. (n) Paddy.

TABLE XII
COMPARISON OF OPERATIONAL EFFICIENCY BETWEEN DIFFERENT DL-BASED MODELS

Metric	HyperspecAE	CyCU	MiSiCNet	HSU
Params	8541	29,873	1,558,617	4,982,234
Flops	5,367	534,424,400	14,078,241,900	824,599,525
Metric	IDNet	SWCNet	SSAFNet	Proposed
Params	538,023	942,240	359,860	5,635,935
Flops	4,832,129,400	8,510,177,900	3,053,798,599	3,475,458,625

preserves local structural continuity while reducing boundary confusion. On the public datasets, the same tendency can also be observed in regions with abrupt abundance variation or fine-scale material transitions. These results indicate that the proposed multiscale spatial aggregation and spatial-spectrum agent attention indeed contribute to more accurate local abundance reconstruction.

G. Efficiency Analysis

This section analyzes the proposed model's computational efficiency against seven methods via parameter count (Params) and FLOPs (see Table XII). The model has 5.64-M Params and 3.48-B FLOPs. In a word, the computational efficiency of MSSANet is higher than lightweight HyperspecAE/CyCU but lower than heavyweight HSU/SWCNet, and balanced versus IDNet/SSAFNet. Though not the most efficient, it achieves a strong tradeoff: outperforming low-complexity models in accuracy while being more efficient than high-overhead competitors. This balance avoids the efficiency-performance dilemma, suiting real-world hyperspectral unmixing needs.

IV. CONCLUSION

In this work, we addressed the challenge of hyperspectral unmixing in complex crop scenes, where mixed pixels frequently occur along field boundaries and in regions with heterogeneous land-cover types. We proposed MS²ANet, which combines a multiscale dilated convolution module, a spatial–spectrum agent attention module, and a triplet MLP to jointly exploit local multiscale context, nonlocal spatial relationships, and global spectral correlations. Extensive experiments on two crop datasets and two public benchmarks demonstrate that MS²ANet consistently improves both abundance and endmember estimation accuracy over recent deep unmixing networks.

Despite these promising results, our method still has some limitations. First, MS²ANet is more computationally demanding than traditional linear unmixing models, and further work is needed to design lighter-weight agent attention modules for large-scale applications. Second, all crop experiments in this article are conducted on UAV imagery; evaluating the proposed network on wide-area airborne or satellite data remains an open problem. In future work, we plan to integrate MS²ANet into an operational UAV-based crop monitoring pipeline and explore model compression techniques to enable real-time deployment.

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